# Boming Jia

Curriculum Vitae — October 2026

Yau Mathematical Sciences Center, Tsinghua University
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## Research Interests

Geometric representation theory and symplectic geometry. Nilpotent orbits, cotangent affinizations, quiver varieties, Coulomb branches, and the Hikita conjecture.

## Academic Appointment

**Yau Mathematical Sciences Center, Tsinghua University** 2022–present
Postdoctoral Researcher (Shuimu Scholar), Beijing, China.

## Education

**University of Chicago** 2016–2022
Ph.D. in Mathematics, June 2022. Advisor: Victor Ginzburg.

**The Ohio State University** 2012–2016
B.S. in Mathematics, May 2016.

## Honors and Awards

**Outstanding Yiyou Scholar Award** Fall 2022, Spring 2023
Qiuzhen College, Tsinghua University. Teaching and mentoring.

**Shuimu Scholar Program**, Tsinghua University 2022

**Lawrence and Josephine Graves Teaching Prize** 2020
University of Chicago, Department of Mathematics.

## Publications and arXiv Preprints

1. Boming Jia. *A Cotangent Model for the Next-to-Minimal Orbit of $E_7$*. arXiv:2609.29537, 2026.
2. Boming Jia. *Hitchin's Conjecture for $\mathfrak{sl}_n$ in Odd Prime Degree*. arXiv:2609.12118, 2026.
3. Boming Jia. *Cotangent Models of Nilpotent Orbit Closures*. arXiv:2608.17206, 2026.
4. Boming Jia. *Minimal Nilpotent Orbits of type $G_2$, $F_4$ and $E_8$*. arXiv:2608.12985, 2026.
5. Boming Jia and Yu Li. *Minimal Nilpotent Orbits and Toric Varieties*. arXiv:2511.02179, 2025.
6. Boming Jia. *Minimal Nilpotent Orbits of type D and E*. arXiv:2501.12406, 2025.
7. Boming Jia. *Highest Weight Varieties and Narayana Numbers*. arXiv:2310.14302, 2023, revised 2026.
8. Boming Jia. *The Affine Closure of $T^*(\mathrm{SL}_n/U)$*.
   *Journal of Lie Theory* **35** (2025), no. 1, 83–100. arXiv:2112.08649, 2021.

## Teaching and Mentoring

**Qiuzhen College, Tsinghua University**
Yiyou Scholar. Undergraduate mentoring and academic support. 2022–2023
Teaching Assistant (TA). Algebra courses. 2023–2026

**University of Chicago** 2018–2022
Instructor. Calculus II, Calculus III, and Linear Algebra.

**University of Chicago REU** 2017
Undergraduate research mentor.

## Selected Talks

**Minimal Nilpotent Orbits and Cluster Toric Varieties** Sept. 2026
Algebra Seminar, BICMR, Peking University, Beijing.

**The Minimal Nilpotent Orbit of $E_6$** Apr. 2026
Workshop on Symplectic Varieties and Projective Geometry,
Morningside Center of Mathematics, Chinese Academy of Sciences, Beijing.

**Minimal Nilpotent Orbits and Cluster Toric Varieties** Nov. 2025
Representation Theory Seminar, AMSS, Chinese Academy of Sciences, Beijing.

**Grassmannians, Nilpotent Orbits, and Toric Varieties** Sept. 2025
Postdoc Workshop V, YMSC, Tsinghua University, Beijing.

**Minimal Nilpotent Orbits and Cluster Toric Varieties** Aug. 2025
Around Singularities in Poisson Geometry,
IASM–BIRS Workshop, Zhejiang University, Hangzhou.

**Minimal Nilpotent Orbits of Type $D$ and $E$** Jan. 2025
Conference on Poisson Geometry and Cluster Algebra,
Chern Institute of Mathematics, Nankai University, Tianjin.

**Minimal Nilpotent Orbit of Type $D$** Dec. 2024
Algebraic Geometry Seminar, AMSS, Chinese Academy of Sciences, Beijing.

**The Affinization of $T^*(G/U)$ and Coulomb Branch** Sept. 2023
Coulomb Branch Student Seminar, YMSC, Tsinghua University, Beijing.

**The Geometry of the Affine Closure of $T^*(G/U)$** Oct. 2022
Workshop on Representation Theory, Fudan University.

**The Geometry of the Affine Closure of $T^*(G/U)$** Aug. 2022
Geometric Representation Theory Workshop, YMSC, Tsinghua University.

**Geometry of the Affine Closure of $T^*(\mathrm{SL}_n/U)$** Jan. 2022
Algebraic Geometry and Commutative Algebra Seminar, University of Notre Dame.

**Geometry of the Affine Closure of $T^*(\mathrm{SL}_n/U)$** Sept. 2021
Symplectic Geometry and Mathematical Physics Seminar,
BICMR, Peking University, Beijing.

## Academic Service

**Organizer, Coulomb Branch Student Seminar** 2023–2024
Yau Mathematical Sciences Center, Tsinghua University.
A seminar for senior undergraduate and beginning graduate students.